

VIGNESH V

Cloud & DevOps Engineer | AWS | Linux | Kubernetes | Docker | Terraform | CI/CD
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PROFESSIONAL SUMMARY

Results-driven Cloud & DevOps Engineer with hands-on expertise in AWS infrastructure automation, Kubernetes orchestration, GitOps, observability, DevSecOps, and multi-cloud Terraform automation. Proven ability to design and deploy scalable, production-ready cloud-native systems with security-first CI/CD pipelines. Eager to contribute to a dynamic engineering team.

TECHNICAL SKILLS

Cloud Platform: AWS (EC2, S3, IAM, VPC, EKS,ECS, ECR, Fargate, CloudWatch, Cost Explorer),GCP

Container & Orchestration: Docker, Kubernetes (K8s), Helm, ECS, Fargate

Infrastructure as Code: Terraform, Ansible

CI/CD : GitHub Actions, Jenkins

Observability: Prometheus, Grafana

Languages: Python, Bash/Shell Scripting

Databases: MySQL, PostgreSQL

Developer Tools: VS Code, Linux(User Management, LVM, SSH, Cron, Systemd, Networking, File Permissions), AWS Console, Git

PROFESSIONAL EXPERIENCE

Freelance DevOps Engineer

Self-Employed | Jul 2024 – Present

- Designed and deployed scalable AWS cloud infrastructure for startups and mid-sized businesses using Terraform and Ansible.
- Built and maintained CI/CD pipelines with GitHub Actions and Jenkins, enabling faster and reliable software delivery.
- Containerized applications using Docker and orchestrated production workloads on Kubernetes (Amazon EKS).
- Implemented monitoring, logging, and alerting solutions with Prometheus, Grafana, and Loki to improve system observability.
- Integrated DevSecOps practices using Trivy, HashiCorp Vault, and automated security scanning into CI/CD workflows.
- Automated infrastructure provisioning, configuration management, and operational tasks using Bash and Python.
- Managed Linux servers, networking, IAM, DNS, SSL/TLS, and cloud security following industry best practices.
- Collaborated with startups and mid-sized companies to modernize cloud infrastructure, optimize deployments, and improve application reliability and scalability.

PROJECTS

GitOps-Driven Microservices Deployment with ArgoCD on AWS EKS

Tech Stack: Kubernetes, AWS EKS, ArgoCD, Helm, Terraform, GitHub Actions, Docker, NGINX Ingress

GitHub: <https://github.com/lnxanae/gitops-argocd-eks.git>

- Implemented fully GitOps-based CD using ArgoCD on AWS EKS with declarative state management across dev, staging, and production environments.
- Configured ArgoCD ApplicationSets and Helm chart repos per environment, eliminating manual kubectl deployments and achieving auto-sync on Git commits.
- Set up ArgoCD RBAC with SSO, enforcing least-privilege access for developers and ops roles across namespaces.
- Integrated sync waves, health checks, and rollback policies for zero-downtime deployments and automatic drift self-healing.
- Provisioned EKS cluster, VPC, and IAM roles via Terraform; GitHub Actions pipeline handles Docker build, ECR push, and Helm chart version bump to trigger ArgoCD sync.
- Reduced mean time to deploy by 60% and eliminated environment drift incidents by enforcing Git as single source of truth.

Full-Stack Observability Platform – Prometheus, Grafana & Loki on Kubernetes

Tech Stack: *Prometheus, Grafana, Loki, Alertmanager, Kubernetes, Helm, Python, AWS CloudWatch, Node Exporter*

GitHub: <https://github.com/lnxanae/k8s-observability-stack.git>

- Deployed production-grade observability stack on Kubernetes using kube-prometheus-stack Helm chart, collecting metrics from 20+ microservices and infrastructure nodes.
- Configured Prometheus scrape configs and custom recording rules to track SLI/SLO metrics: request latency (p95, p99), error rates, and pod resource saturation.
- Built 10+ Grafana dashboards covering cluster health, application performance, and AWS cost metrics with variable-driven multi-environment visibility.
- Integrated Loki + Promtail for centralized log aggregation from all Kubernetes pods, enabling correlated log-metric investigation from a single Grafana pane.
- Configured Alertmanager routing with severity tiers, delivering critical alerts to Slack and PagerDuty with embedded runbook links.

DevSecOps Pipeline – Shift-Left Security with Vault, Trivy & OPA

Tech Stack: *HashiCorp Vault, Trivy, OPA/Gatekeeper, Semgrep, GitHub Actions, Docker, AWS EKS, Terraform*

GitHub: <https://github.com/lnxanae/devsecops-pipeline.git>

- Designed shift-left security pipeline integrating SAST (Semgrep), container image scanning (Trivy), and Kubernetes admission policy enforcement (OPA Gatekeeper) at every CI/CD stage.
- Deployed HashiCorp Vault on Kubernetes using Vault Agent Injector, eliminating hardcoded secrets; automated AWS IAM credential rotation every 24 hours.
- Enforced OPA Gatekeeper admission policies blocking privileged containers, missing resource limits, and images from unverified registries.
- Built GitHub Actions security gate failing builds on CRITICAL/HIGH CVEs from Trivy, with weekly baseline reports sent to the security team.
- Implemented Kubernetes RBAC audit logging shipped to S3 for compliance reporting aligned with SOC 2 access control requirements.

Multi-Cloud Infrastructure Automation – AWS & GCP with Terraform

Tech Stack: *Terraform, AWS (EKS, RDS, S3, Route53), GCP (GKE, Cloud SQL), GitHub Actions, Python, HashiCorp Vault*

GitHub: <https://github.com/lnxanae/multi-cloud-terraform.git>

- Architected multi-cloud infrastructure spanning AWS and GCP using modular Terraform, provisioning parallel EKS and GKE clusters for workload distribution and vendor resilience.
- Designed Terraform module hierarchy with environment-specific tfvars (dev/staging/prod) and remote state backends on AWS S3 and GCS for consistent cross-cloud deployments.
- Configured AWS Route 53 latency-based routing and GCP Cloud DNS to distribute traffic between clouds with sub-100ms failover.
- Automated cross-cloud IAM credential management using HashiCorp Vault dynamic secrets, eliminating static long-lived credentials in both clouds.
- Built unified GitHub Actions pipeline using matrix builds to plan and apply Terraform changes for AWS and GCP in parallel, reducing infrastructure update time by 55%.

EDUCATION

RVS College of Engineering (Anna University, Chennai) | Sept 2021 – May 2025

B.E. Computer Science and Engineering | CGPA: 8.6 / 10

Relevant Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Cloud Computing, DevOps

CERTIFICATIONS

- Oracle Cloud Infrastructure 2023 Certified Foundations Associate – Sept 2023
- High Performance Computing Technology (Career Dot) – Jun 2024
- Network Essentials (CISCO) – Aug 2022

KEY STRENGTHS

- Deployed GitOps pipelines (ArgoCD), full-stack observability (Prometheus/Grafana/Loki), and DevSecOps automation (Vault/Trivy/OPA) on production Kubernetes clusters.

- Designed multi-cloud Terraform automation across AWS and GCP with unified CI/CD and dynamic secrets management.
- Strong problem-solving skills; built AI-based cloud cost monitoring with anomaly detection and forecasting using Prophet.
- CGPA 8.6 – maintained academic excellence alongside intensive hands-on project development.